SYSTEM DOCUMENTATION LOG

PROJECT HIND-KE-SITARE

ORBITAL COMMUNICATIONS PROTOCOL

LEAD COMMANDERS

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MISSION APPROACH | PROBLEM RESOLUTION

To build the CPaaS voice automation system, we designed a real-time modular architecture that directly integrates telephony, speech processing, and contextual AI into a low-latency pipeline. We used Asterisk ARI with ExternalMedia to gain full programmatic control over the call lifecycle and RTP media streams. This allows the bot to receive and send real RTP audio instead of relying on black-box telephony APIs. ExternalMedia enables isolated mixing bridges for safe cleanup and concurrent call handling. For testing and SIP interaction, we used Zoiper as a softphone. **Additionally, we implemented an intelligent outbound calling capability. The system proactively initiates calls to follow up on conversations from prior days (data of such users can be stored in database, we hardcoded the info of the user), to verify pending requests and seek user confirmation.** For speech processing, we selected faster-whisper for local speech-to-text transcription with beam_size=5 to improve accuracy in noisy telephony audio. GPU acceleration is used when available to reduce latency. For text-to-speech, we used the Piper neural TTS model because it provides low-latency, controllable output and efficient sample-rate handling. For conversational intelligence, we integrated Google Gemini (streaming mode). Conversation history is managed manually to support interruption handling and automatic MoM generation. Using local ASR and TTS reduces cloud cost and privacy exposure, while the streaming LLM provides contextual reasoning and natural interaction.

The system operates using a state machine (LISTENING, THINKING, SPEAKING) to manage real-time behavior. Grace periods prevent unstable transitions. Speech boundaries are detected using Silero VAD, and audio is preprocessed using a 4th-order bandpass filter (80–7000 Hz) with gain amplification. Incoming 8 kHz telephony audio is resampled to 16 kHz for Whisper and VAD.

To handle natural interruptions, we implemented a dynamic echo-gating barge-in algorithm. It continuously monitors incoming audio energy against a dynamic threshold based on the bot's current output energy. If the energy breaches this gate and VAD detects human speech for a sustained duration (4 chunks, $\sim$125 ms), the system instantly halts the TTS and transitions to listening. This staged approach minimizes false interruptions while maintaining rapid reaction latency.

Responses are streamed from the LLM and chunked into sentence-level segments, which are immediately forwarded to TTS for playback. Streaming reduces perceived delay and allows the bot to sound more natural. A closing protocol uses a [HANGUP] token to terminate calls professionally.

When a call ends (StasisEnd), the system triggers generate_mom() using the stored conversation history. A focused summarization prompt generates a structured Minutes of Meeting highlight-

ing requirements, budget, location, key discussion points, and action items, which is saved to minutes_of_meeting.txt.

Overall, we implemented a scalable, low-latency, and cost-efficient AI voice architecture with full telephony control, intelligent interruption handling, streaming responses, GPU-accelerated transcription, and automated business summarization.

TECHNOLOGY PAYLOAD | JUSTIFICATION

[Asterisk (ARI + ExternalMedia)]

- Open-source, production-grade PBX.
- Full programmatic control using ARI (Asterisk REST Interface).
- ExternalMedia allows direct RTP streaming to our AI engine.
- Supports SIP, PJSIP, and enterprise telephony standards.

[Faster-Whisper (Local Whisper Model)]

- Runs locally (privacy-preserving), no api limits.
- GPU acceleration support via CTranslate2.
- Good balance between latency and accuracy.
- Works well with telephony audio (8kHz $\rightarrow$ resampled 16kHz).

[Piper (Local Neural TTS)]

- Fully local inference.
- Very low latency.
- Small model size.
- Easy RTP integration.

[Gemini Flash (Streaming)]

- Fast response streaming.
- Optimized for low latency.
- Good reasoning ability.
- Contextually aware.
- API is free.

[Silero VAD (ONNX Runtime)]

- Lightweight and fast.
- High accuracy for telephony audio.
- ONNX runtime makes deployment easy.
- Supports stateful LSTM inference.
- Works reliably at 8kHz and 16kHz.

» ADVANTAGES OVER ALTERNATIVE SOLUTIONS

- Asterisk over Twilio, because it is open sourced and twilio is not free.

- Faster-whisper is an open-source, optimized reimplementation of OpenAI's Whisper (using CTranslate2) designed for efficient self-hosting. No other alternatives were efficient for self hosting on a local computer and cost effective.
- Piper has a very low latency. Piper was used because of the improved latency and performance.
- Gemini api has the best trade-off between performance and cost. Other API keys were not free.
- WebRTC VAD: less accurate in noisy environment. Deep neural VAD (heavy models): can increase latency.

ORBITAL MECHANICS | TECHNICAL IMPLEMENTATION

» 1. End-to-End Telemetry Flow (Execution Pipeline)

When a call arrives (Inbound):

- Asterisk ARI triggers StasisStart
- Channel is answered programmatically.
- A dedicated bridge is created.
- ExternalMedia is attached to stream RTP audio to our VoiceBot (UDP 4000).
- VoiceBot begins RTP receive loop.
- Bot plays greeting using TTS.
- Real-time conversation begins.

Outbound Workflow:

- Asterisk ARI originates a new call to the user.
- The call is receiver in the Zoiper softphone.
- Upon answer, the LLM is seeded in the system prompt to ask for and sharing more info to the user.
- The bot asks for confirmation regarding the previously made request.

Pipeline per user turn:

- RTP → Audio Preprocessing → VAD → Buffer → Whisper STT → LLM Stream → TTS Stream → RTP Back

When call ends:

- StasisEnd event triggers MoM generation in a background thread.
- Bridges and RTP channels are cleaned.
- State resets.

» 2. Acoustic Signal Processing

Incoming audio from Asterisk:

- Format: 8kHz 16-bit PCM (SLIN)
- Converted from Big-endian to Little-endian
- Resampled: 8kHz → 16kHz (Whisper requirement)

Chunking:

- CHUNK_SIZE = 320 samples @ 16kHz
- 320 samples = 20ms per frame
- Processing is done frame-by-frame for low latency

Audio preprocessing:

- 4th order bandpass filter: 80Hz – 7000Hz
- Digital gain multiplier: 3.0
- Helps improve speech clarity before ASR

» 3. Voice Activity Detection (VAD) Model: Silero VAD (ONNX runtime)

Key Parameters:

- Threshold: 0.75
- Min silence duration: 600 ms
- Speech pad: 30 ms

Speech Start Detection If VAD probability $\geq 0.75 \rightarrow$ Speech starts.
Speech End Detection Speech is considered ended when:

- Probability drops below (threshold - 0.15)
- Silence persists for $\geq$ 600 ms

This prevents premature cutoff during natural pauses.

» 4. Core State Machine Three states control the bot:

- LISTENING
- THINKING
- SPEAKING

LISTENING

- Collect audio while VAD is triggered.
- When VAD detects speech end $\rightarrow$ move to THINKING.

THINKING

- Transcription occurs.
- LLM streaming begins.
- When first chunk arrives $\rightarrow$ transition to SPEAKING.

SPEAKING

- TTS output is streamed via RTP.
- Bot monitors for barge-in.

» 5. Pause Timing & Grace Windows These values control smooth interaction:

Parameter	Value	Purpose
CHUNK_SIZE	20ms	Audio frame resolution
Min Silence	600ms	End-of-speech detection
State Grace	400ms	Prevent false barge immediately after speaking
Post Grace	1000ms	Wait after TTS finishes before listening
Barge Thresh	4 chunks	∼125ms stable speech required

» 6. Collision Avoidance (Barge-In Detection)

Barge-in (interruption) is handled only when the bot is in the SPEAKING state, after the state transition grace period has passed.

Conditions required for interruption:

- **Dynamic Echo Gate:** The system calculates a dynamic energy threshold based on the synthesizer's current output: $\max(2000, 0.8 \times \text{bot output energy})$.
- **Energy Verification:** The mean energy of the incoming audio frame must exceed this dynamic threshold to be considered a candidate.
- **VAD Confirmation:** The Silero VAD probability must exceed the defined threshold.
- **Persistence:** The incoming speech must persist for ≥ 4 consecutive chunks (4×20ms = ∼125ms of stable speech).

If all conditions are satisfied:

- `synthesizer.stop()` is executed to halt audio playback.
- The buffered `pre_barge_buffer` is captured to ensure no user speech is lost.
- State transitions instantly to `LISTENING`.
- The bot yields the channel back to the user.

» 7. Where Streaming Happens

1. **LLM Streaming**

- Location: brain.generate_response_stream()
- Uses stream=True
- Text arrives in chunks
- Each chunk appended to sentence buffer
- As soon as a sentence ends → sent to TTS immediately
- This reduces perceived latency significantly.

2. **TTS Streaming**

- Location: synthesizer.synthesize_text()
- Piper generates audio chunks
- Each chunk is resampled to 8kHz
- Put into playback queue
- RTP packets sent every 20ms
- TTS playback worker sends RTP frames continuously.

» 8. RTP Timing Control In _playback_worker():

- FRAME_BYTES = 320 bytes
- FRAME_DURATION = 20ms
- RTP timestamp incremented by samples sent
- Marker bit used on first packet of each talkspurt
- This ensures telecom-grade timing accuracy.

» 9. Hardware Acceleration (GPU) GPU is used in Whisper STT.

Detection:

```
import ctranslate2
if ctranslate2.get_cuda_device_count() > 0:
    device = "cuda"
    compute_type = "float16"
```

If GPU available:

- Whisper runs on CUDA
- Faster inference
- Lower transcription latency

If no GPU:

- Falls back to CPU with int8 quantization

No GPU is used for:

- TTS (Piper runs CPU)
- VAD (ONNX CPU)
- LLM (cloud API)

» 10. Post-Flight Report (MoM Generation) Triggered when:

- ARI → StasisEnd

Process:

- Conversation history stored in Brain
- Special MoM prompt appended
- Gemini generates structured summary
- Saved to minutes_of_meeting.txt
- This runs in a background thread so telephony cleanup is not blocked.

» 11. Latency Breakdown (Typical) Approximate timings (GPU enabled):

Stage	Latency
VAD Detection	∼20ms
Whisper ASR	300–600ms
LLM First Token	200–500ms
TTS First Packet	<50ms
Total Perceived Delay	∼700ms–1 s

Streaming reduces perceived wait time.

» 12. Failure Handling & Stability

- ☆ ARI startup cleanup removes orphan bridges.
- ☆ JSON body used instead of query params (avoids Asterisk parsing crash).
- ☆ Locking ensures one setup per channel.
- ☆ Graceful synthesizer abort via abort_event.
- ☆ Debug WAV files saved for ASR validation.

CONSTELLATION ARCHITECTURE | HIGH LEVEL DESIGN

The system follows a modular, layered architecture separating telephony, real-time voice processing, conversational intelligence, and post-call analytics. At the telephony layer, Asterisk (via ARI and ExternalMedia) manages call lifecycle, SIP signaling, and RTP streaming. It forwards raw RTP audio directly to the AI engine and delivers synthesized audio back to the caller, giving full programmatic control over calls. The real-time AI engine handles audio preprocessing, voice activity detection, speech-to-text transcription, LLM-based response generation (streaming), and text-to-speech synthesis. A state machine (LISTENING, THINKING, SPEAKING) manages interaction flow and interruption handling. Conversation history is maintained throughout the call and reused in the post-call layer to automatically generate structured Minutes of Meeting (MoM). This layered, streaming architecture ensures low latency, scalability, modularity, and enterprise-grade control over the complete call workflow.

» Constellation Workflow (End-to-End).

```
                    +-------------------------+
                    |        Caller           |
                    | (Customer Phone / SIP)  |
                    +------------+------------+
                                 |
                                 V
                    +-------------------------+
                    |        Asterisk         |
                    |  (PBX + ARI + Bridges)  |
                    +------------+------------+
                                 |
                    RTP (ExternalMedia)
                                 |
                                 V
+-------------------------------------------------+
```

```
            |                Command Module                 |
            |            (Real-Time AI Call Engine)         |
            +-----------------------------------------------+
                                |
      +-----------------------------+-----------------------------+
      |                             |                             |
      V                             V                             V
+------------+                +-------------+                +--------------+
| Audio      |                | VAD (Silero)|                | Call State   |
| Preproc    |                | Speech Det. |                | Manager      |
| (Filter/   |                |             |                |(Listen/Think/|
| Gain)      |                +-------------+                | Speak)       |
+------------+                       |                       +--------------+
                                     V
                              +-------------+
                              | Transcriber |
                              |(Whisper STT)|
                              +-------------+
                                     |
                                     V
                              +-------------+
                              |    Brain    |
                              |(LLM-Gemini) |
                              |Context Logic|
                              +-------------+
                                     |
                                     V
                              +-------------+
                              | Synthesizer |
                              | (Piper TTS) |
                              +-------------+
                                     |
                                     V
                          RTP Audio Back to Asterisk
                                     |
                                     V
                                  Caller
            Call Ends (StasisEnd Event)
                        |
                        V
                +--------------+
                | Conversation |
                | History Buf  |
                +--------------+
                        |
                        V
                +--------------+
                | Brain (LLM)  |
                | MoM Generator|
                +--------------+
                        |
                        V
                +--------------+
                | minutes_of_  |
                | meeting.txt  |
                +--------------+
```

1. Caller speaks
2. RTP packet → Asterisk → ExternalMedia
3. RTP → VoiceBot (UDP socket)
4. Audio preprocessing (filter + gain)
5. VAD detects speech start/end

6. Complete speech chunk → Whisper ASR
7. Transcribed text → Brain (LLM)
8. LLM streams response text
9. Text chunks → Piper TTS
10. TTS → RTP packets → Asterisk
11. Caller hears response
12. On hangup → Generate MoM